



Crime hotspot analysis and mapping using geospatial technology in Dessie city, Ethiopia

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ABSTRACT

The problem of urban crime in most developing regions has increased, and spatial configurations tend to be consistent with social and environmental dynamics. The paper has explored the spatial concentration and distribution of crime in Dessie City, Ethiopia, through geospatial technology. The study combined police records data on crime with Geographic Information System (GIS) data to determine locations in which criminal activity is likely to be concentrated, and also the social circumstances of these places. The results indicated that the core areas of *Hote*, *Arada*, and *Segno* have more crime concentration, with the North and peripheral areas having quite low crime levels. The findings also revealed that theft and intimidation are most prevalent, most of them being committed in the place of prominent places and business hubs, but murder is not that common, but geographically distinct. Directional and temporal analysis showed that the majority of crimes are committed during the day and are more likely to have a North-South spatial pattern. The paper emphasizes the role of geographic analysis in enhancing the knowledge on the interrelation of crime and city form, which can be useful in directed policing and the allocation of resources. The knowledge derived can help in criminological studies as well as in practical ways of making Ethiopian cities with a high growth rate safer and have efficient ways of governing the city.

1. Introduction

The rise of crime and delinquency remains a major concern across both industrialized and developing nations. Over the past three decades, police statistics have reported dramatic increases in crime in many parts of the world. The International Crime Victim Survey (ICVS), which covers six major geographic zones-Africa, Asia, Central and Eastern Europe, Latin America, and Western Europe-found that more than half of urban residents experienced crime at least once between 1989 and 1996 [1]. Crime manifests in diverse forms, including homicide, assault, theft, and fraud [2]. However, significant regional variations exist: while violent crime has declined in Western Europe and North America, it remains persistently high in Latin America and sub-Saharan Africa, as noted in the Journal of Quantitative Criminology [3]. Ethiopia, situated in sub-Saharan Africa, requires closer monitoring, as rapid socioeconomic changes have contributed to rising crime rates [4]. These challenges call for innovative approaches to crime data management, analysis, and visualization to support more effective resource allocation [5]. Geographic Information Systems (GIS) provide essential tools for

this purpose, enabling researchers to map and analyze crime by linking spatial and social characteristics [6]. Such visualizations reveal hidden patterns and support law enforcement strategies [7].

Crime imposes severe social and economic costs, including fear, loss of life and property, reduced income, displacement, and the diversion of development funds toward security expenditures [8]. In Ethiopia, data from the Federal Police Commission reveal a sharp rise in criminal incidents that undermine community well-being. Yet, crime analysis remains largely manual, with little use of spatial data for prevention or policing [9]. The Amhara Police Commission (2020) reported approximately 20,000 criminal cases in a single year, reflecting a 3.7 % increase from 2019. Major categories included theft, fraud, homicide, burglary, and looting, though such reports lack spatial context. Studies also highlight that socioeconomic factors such as poverty, unemployment, and demographic shifts play a decisive role in shaping crime patterns [10,11]. Urban crime has become a global phenomenon, strongly influenced by inequality, political instability, and rapid urbanization [12,13]. In Ethiopia, particularly in cities like Dessie, recent socio-demographic changes have intensified criminal activity [14,15].

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Dessie's police department still relies on non-computerized record-keeping, limiting its ability to manage crime data efficiently or respond to emerging hotspots.

Crime has inherent spatial dimensions, defined by location, timing, and process. Effective policing requires accurate geographic information about crime-prone areas [16–18]. Previous studies in Ethiopia, such as analyses conducted in Addis Ababa's Lideta sub-city, provided useful insights but failed to identify clear hotspots or spatial correlations [19]. Dessie City offers a representative case for further study due to its rapid urban growth and diverse population [20,21]. The city's population expansion and economic transition create both opportunities and vulnerabilities, making it an important context for spatial crime research. This study is also guided by criminological theories that explain why crime concentrates in specific places. Routine activity theory [22] argues that crime occurs where motivated offenders encounter suitable targets in the absence of capable guardians. Social disorganization theory [15] emphasizes that poverty, residential instability, and weak community ties reduce informal social control and increase crime risk. Environmental criminology [23], highlights how land use, transport hubs, and public spaces shape opportunities for crime. By integrating these theoretical perspectives with advanced geospatial methods, this study contributes to a more comprehensive understanding of urban crime in Dessie City and provides evidence to inform local law enforcement strategies.

Recent criminological research highlights that crime is rarely random but is instead embedded in the broader socio-economic and spatial fabric of cities [24], formulated the "law of crime concentration," demonstrating that a small proportion of places typically account for the majority of criminal incidents. This principle has been confirmed in both Global North and Global South contexts, including rapidly urbanizing African and Latin American cities [25,26]. In Ethiopia, emerging studies show that inequality, demographic change, and weak urban governance contribute to crime concentration in specific micro-places as stated by [14,27]. Similar work in Latin American cities has emphasized the clustering of crime around markets, transport hubs, and informal economic zones, pointing to the interaction of routine activities and limited guardianship [26]. By situating Dessie within these wider criminological debates, this study strengthens the connection between geospatial methods and crime science, highlighting both local realities and their resonance with global patterns.

2. Methodology of the study

2.1. Description of the study area

This section describes Dessie City in terms of geography, demographics, climate, economy, and historical significance. Dessie lies at an elevation of about 2700 m above sea level, between longitudes 39°35'36" and 39°40'45" east and latitudes 11°4'59" and 11°12'36" north. It is located roughly 400 km north of Addis Ababa along the Mekelle route, in northern Ethiopia. As the capital of South Wollo, Dessie is marked as 'C' on the study area map, with 'A' representing Ethiopia and 'B' the Amhara region shown in tourmaline green. The city's population is estimated at 443,570, with 51.21 % males and 48.79 % females. Amhara is the dominant ethnic group, alongside sizeable Oromo and Tigrayan communities. This diversity fosters rich intercultural and religious interactions, with Muslims and Orthodox Christians forming the majority. Age distribution shows that children (0–14 years) account for about 40 % of the population, the working-age group (15–64 years) about 55 %, and the elderly (65 years and older) around 5 % [28].

Dessie has a temperate climate, with a rainy season from June to September and a dry season from October to May. Temperatures range from 10 °C at night to about 25 °C during the day [29]. These conditions support agriculture, the backbone of the local economy. Major crops include cereals (maize, wheat, and teff), pulses (beans and chickpeas), and cash crops (onions, tomatoes, and peppers). Most households

practice subsistence farming, and income levels vary widely, with a large portion of residents living below the poverty line. Agriculture, trade, and small-scale industries form the main employment sectors, although services are growing [30]. Access to utilities remains limited, with many residents lacking reliable electricity and safe water. Living conditions are further shaped by educational opportunities. The literacy rate is about 70 %, but disparities persist between urban and rural areas. Primary school enrolment is relatively high, though dropout rates increase in higher grades due to economic pressures. Access to secondary education has improved, but only a small share of students advance to tertiary education [31–33].

Historically, Dessie has served as a commercial hub since the late 19th century and played strategic roles during political and military events in Ethiopia, such as the Italian invasion of the 1930s and resistance against the Derg regime [34]. Crime rates began to rise in the early 2000s, coinciding with urbanization and socio-economic changes. Federal Police records show increases in theft, assault, and homicide over the past decade, consistent with patterns observed in other Ethiopian cities. High-profile incidents, including gang-related violence, illustrate evolving forms of criminal activity. Social dynamics also shape crime: high youth populations, unemployment, and poverty contribute to criminality, while inadequate education further aggravates the problem [35,36].

2.2. Research design

This study applied both quantitative and qualitative approaches. Quantitative methods were used to process field data with GIS software and algorithms, producing maps and numerical outputs. Qualitative methods involved interpreting and explaining these outcomes in narrative form. The findings from both approaches were integrated, with qualitative insights reinforcing the quantitative evidence. Accordingly, a concurrent mixed research design was adopted, enabling the simultaneous collection and analysis of both qualitative and quantitative data. This strategy allowed reciprocal complementarity, improving the robustness of the analysis by combining numerical precision with contextual interpretation.

2.3. Data types and sources

Primary data were collected using a Garmin GPS device to record spatial points for police stations and frequently occurring crime scenes. Relevant characteristics of crimes—such as type, occurrence, victims' age, number of victims, and time of occurrence in each sub-city from 2018 to 2021—were obtained through interviews with sub-city police stations and from their official documents and reports as secondary data. While police records and interviews provide valuable insights into local crime dynamics, they have limitations. Crime data in Ethiopia are often recorded manually, which can result in underreporting or inconsistencies across sub-cities. GPS-based collection of crime locations depends on the accuracy of police reports and officer recall, potentially introducing reporting bias. Moreover, access to police archives is limited, so the data are not fully replicable by external researchers. To mitigate these concerns, the study triangulated multiple sources—including official reports, police interviews, and direct GPS observations—to enhance reliability. Despite these efforts, some data limitations remain. Underreporting of certain crimes, particularly domestic violence and petty theft, may have led to underestimation in specific neighborhoods. Similarly, reliance on local police interviews introduces potential subjective recall bias. Nevertheless, the strong spatial clustering revealed by Moran's I and hotspot analysis suggests that the broad geographic patterns observed are robust, even if individual counts vary. This finding aligns with comparable African urban studies where police data served as the primary source [29,37]. Regarding consistency over the study period (2018–2021), the police department confirmed that crime categories—such as theft, intimidation,

burglary, and homicide were consistently defined. Although minor changes in reporting procedures occurred (introduction of digital registers in late 2021), the classification of crime types remained stable, ensuring comparability. Additionally, books, publications, and journal articles were used to support the study.

2.4. Methods of data analysis

This study employed a combination of spatial statistical and geostatistical techniques to identify and explain patterns of crime in Dessie City. The analysis was carried out in ArcGIS Pro and supported by criminological theory to ensure that statistical results were interpreted within their social context. First, optimized hotspot analysis (Getis-Ord G_i^*) was applied to locate statistically significant clusters of crime. A high z-score with a low p-value indicated the presence of hotspots, while low negative z-scores revealed cold spots. This technique is widely used in crime mapping because it provides a robust statistical basis for identifying areas with concentrations of high or low criminal activity [38]. Second, spatial autocorrelation was measured using Moran's I, which tests whether crime incidents are distributed randomly, clustered, or dispersed across space. Moran's I is a standard tool in spatial criminology, allowing researchers to assess whether observed crime hotspots reflect genuine spatial dependencies or chance patterns [17,39].

Third, semivariogram modeling was conducted to evaluate the degree of similarity among crime events over distance. This geostatistical method quantifies the range of spatial influence and helps assess whether clustering occurs at local or citywide scales [7,40]. Fourth, Inverse Distance Weighting (IDW) interpolation was used to predict potential future crime-prone areas. IDW assumes that closer points exert stronger influence than distant points, thereby creating a continuous surface of estimated crime intensity [41]. This method has been successfully applied in urban crime prediction, especially where police data are limited [42]. Finally, spatial descriptive statistics were employed to analyze crime directionality and central tendency. The mean center statistic identified the geographic center of all crime types, while the standard deviational ellipse (SDE) captured the spread and directional trend of incidents. These techniques provide insight into whether crime distribution follows systematic directional patterns, which is important for resource allocation and urban safety planning as stated by [43]. The IDW method is expressed as:

$$Z_j = \frac{\sum_{i=1}^n \frac{Z_i}{d_{ij}^n}}{\sum_{i=1}^n \frac{1}{d_{ij}^n}}, n > 0$$

Where, Z_i is the value of the known point, d_{ij} is the distance to a known point, Z_j is the unknown point n is a user-selected exponent.

Geary's C and Moran's I are two widely used methods for evaluating spatial autocorrelation. Moran's I, in particular, is commonly applied to detect clustering of crime incidents [44], stated as:

$$I = \frac{n \sum_{i=1}^n \sum_{j=1}^n w_{ij} z_i z_j}{S_0 \sum_{i=1}^n z_i^2}$$

Where, z_i is the deviation of an attribute for feature i from its mean ($x_i - \bar{x}$), w_{ij} is the spatial weight between features i and j , where n is equal to the total number of features, and S_0 is the aggregate of all spatial weights, which is $\sum_{i=1}^n \sum_{j=1}^n w_{ij}$. The Z_i The score for statistics is computed as: $z_i = \frac{I - E[I]}{\sqrt{V[I]}}$ Where, $[I] = \frac{-1}{(n-1)}$, and $V[I] = E[I^2] - E[I]^2$.

The semi-variogram is defined as:

$$\gamma(s_i, s_j) = \frac{1}{2} \text{var}(Z(s_i) - Z(s_j))$$

Where, var the variance, the anticipation that two places, s_i and s_j , that are close to one another in terms of the distance measure of $d(s_i, s_j)$ will be similar, therefore the difference in their values, $Z(s_j) - Z(s_i)$, will

be minimal. The gap $Z(s_i) - Z(s_j)$ will increase as s_i and s_j grow apart because they become less similar to one another Fig. 1.

The flowchart in Fig. 2 illustrates the methodology employed to study crime patterns in Dessie City. The process begins with the collection of primary data, including GPS point data from local police stations detailing crime incidents and their characteristics from 2018 to 2021, supplemented by secondary sources to provide comprehensive context. The study integrates both quantitative and qualitative approaches, combining statistical analyses with insights from interviews with police officials. Key analytical techniques include semivariogram modeling, spatial autocorrelation analysis using Moran's I, and standard deviation ellipse and mean center statistics to determine the mean center and directional trends of crime. Potential future crime locations were identified using Inverse Distance Weighted (IDW) interpolation, while optimized hotspot statistical methods pinpointed significant crime hotspots and cold spots. The flowchart visually represents these sequential steps, from data collection to analysis, culminating in the identification of statistically significant crime patterns. This structured approach facilitates targeted law enforcement strategies and enhances understanding of the spatial distribution of crime in Dessie City, supporting effective resource allocation and intervention planning.

3. Result

Understanding the types of crimes and their spatial distribution is essential for analyzing crime dynamics. Based on interviews with police professionals, the study documented 810 spatially dispersed incidents from 2018 to 2021, categorized as theft (660 incidents), murder (147), intimidation (315), and other crimes. This detailed classification provides insights into the temporal and spatial patterns of crime in Dessie [45]. In total, 4462 events were linked to these spatial characteristics over the study period, with an annual breakdown of 1122 offenses in 2018, 1106 in 2019, 1107 in 2020, and 1127 in 2021. Fig. 3 shows that crime incidents peaked in 2018, declined slightly in 2019, and then experienced a modest increase in 2020 and 2021. Across all years, theft was the most frequently reported crime, while murder was the least common in each sub-city. Notably, in the Menasha sub-city, most crimes involved deception, with intimidation reported less frequently.

The identified criminal activities include theft, intimidation, looting/beating, duping, burglary, and murder, as illustrated in Fig. 4a. The distribution of incidents across sub-cities was as follows: Arada (197), Hote (191), Segno Gebeya (181), Bounbouwuha (150), and Menafesha (91). Primary crime locations included vacant lots, commercial centers, prominent streetside taxi fermatas, single-family residences, hotels, and market centers. Violent crimes such as murder were mainly concentrated in vacant areas near the Hote fields. Key taxi fermatas involved

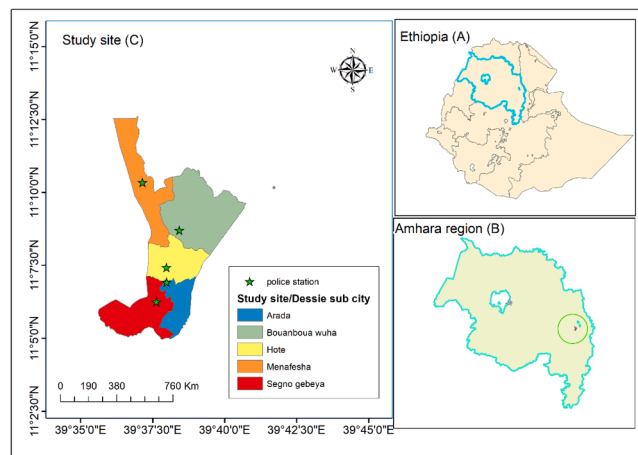


Fig. 1. Location map of the study area.

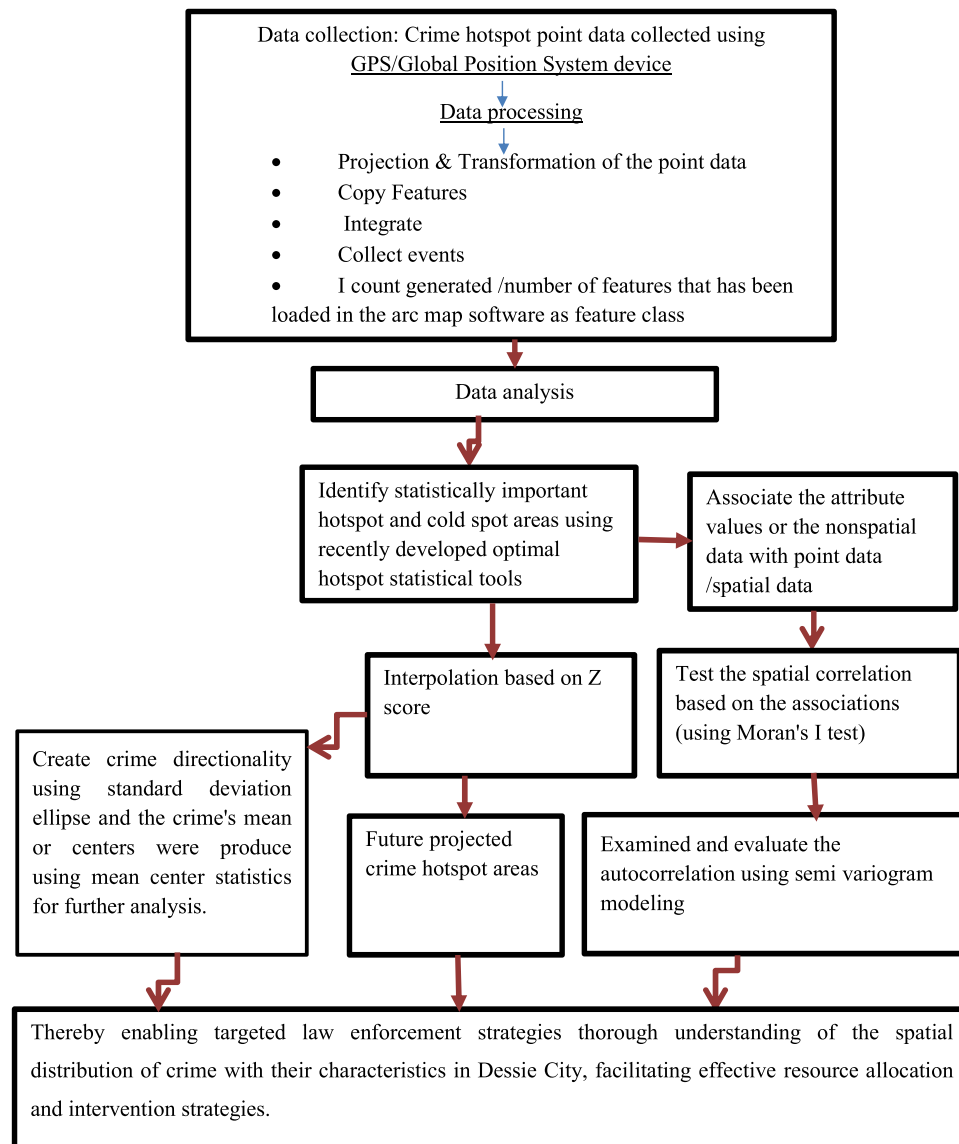


Fig. 2. Flow chart of the study methodology.

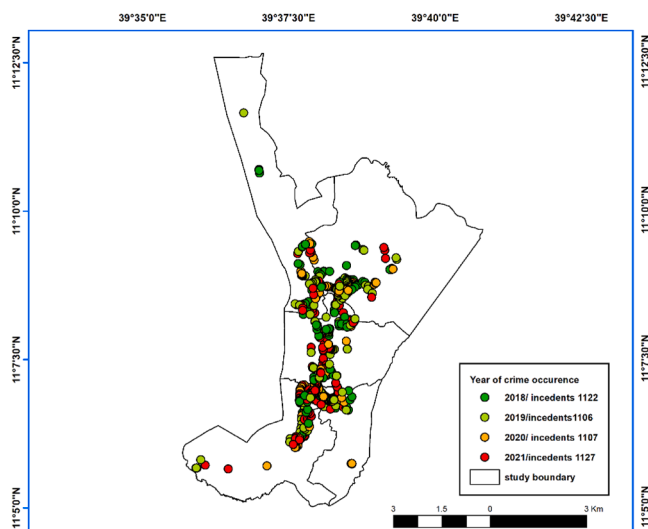


Fig. 3. Temporal distribution of crime in the study area.

included Arada, Paisa, Segno, Meneharia, Shel, Melaku, Menafesha, Bounbouwuha, Dandiboru, and Tekuam, while significant market centers included Segno and Robit Gebeya, and commercial centers such as Arada and Jumuruk.

Crime occurrences were classified on a scale from 1 (lowest) to 28 (highest), with theft being the most frequent and murder the least. Using natural breaks (Jenks) for classification, occurrences ranging from 1 to 5 were shaded light green. The recorded frequencies were as follows: theft (660), looting/beatings (447), duping/deceit (398), intimidation (315), and murder (147). Six to ten murder incidents were highlighted in yellow. Overall, theft was the most prevalent crime (999 incidents), followed by intimidation (483), looting/beatings (88), duping (58), and burglary (45). In the 11–16 incident range (faintly red), theft remained the most observed offense (235), with intimidation second (188). For the 17–28 incident range (tinted red), theft was again the most frequent (216), followed by intimidation (156), as shown in Fig. 4b

In temporal crime hotspots, identifying the timing of criminal acts is as important as determining their locations. The results indicate that theft was the most common crime, accounting for 47.29 % of all offenses. Overall, 43.14 % of incidents occurred during the day, while 4.15 % occurred at night. Theft incidents specifically were more frequent during the day, representing 27 % of cases, compared with 13 % at

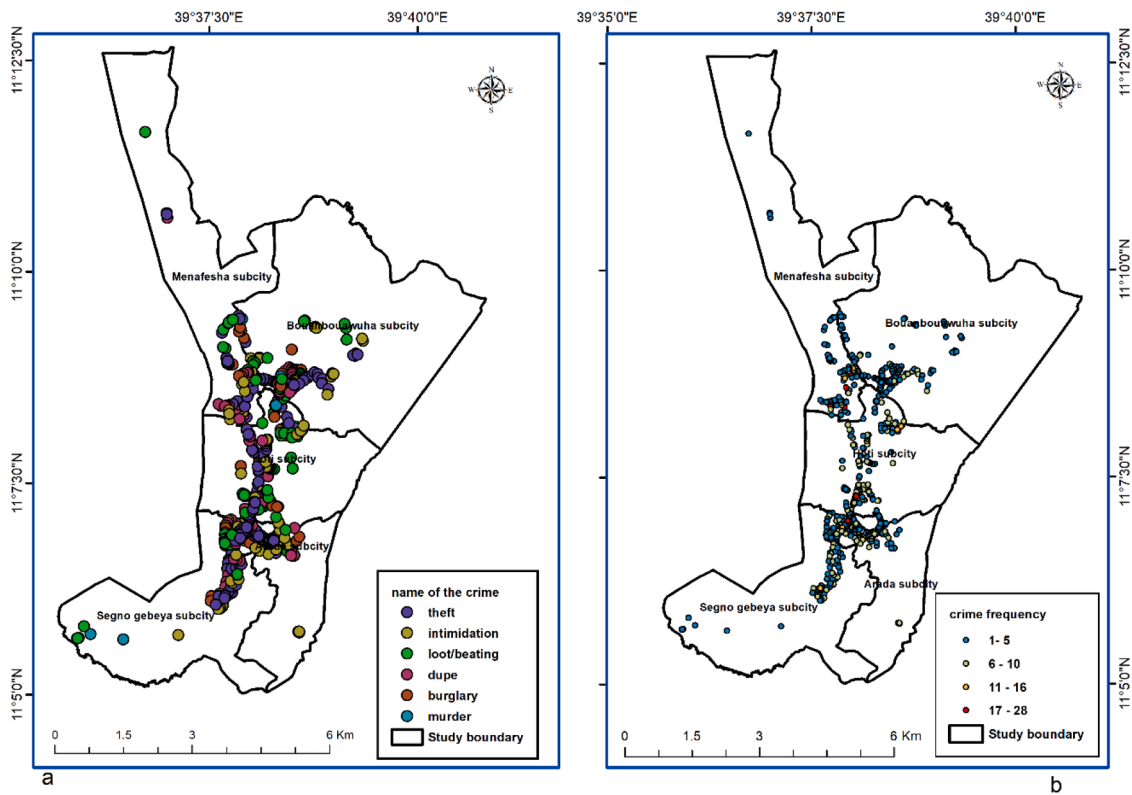


Fig. 4. Distribution(a) and occurrence (b) of crime in dessie city 2018–2021.

night. Incidents of intimidation and theft together accounted for 12.69 % of cases, with 6.91 % occurring during the day and 5.09 % at night. Deception-related crimes accounted for 10.22 % during the day, 7.15 %

at night, and 3.07 % in other time intervals. Burglaries represented 4.30 % of incidents, with 2.24 % occurring during the day and 2.06 % at night. Murders, constituting 0.61 % of all cases, occurred both during

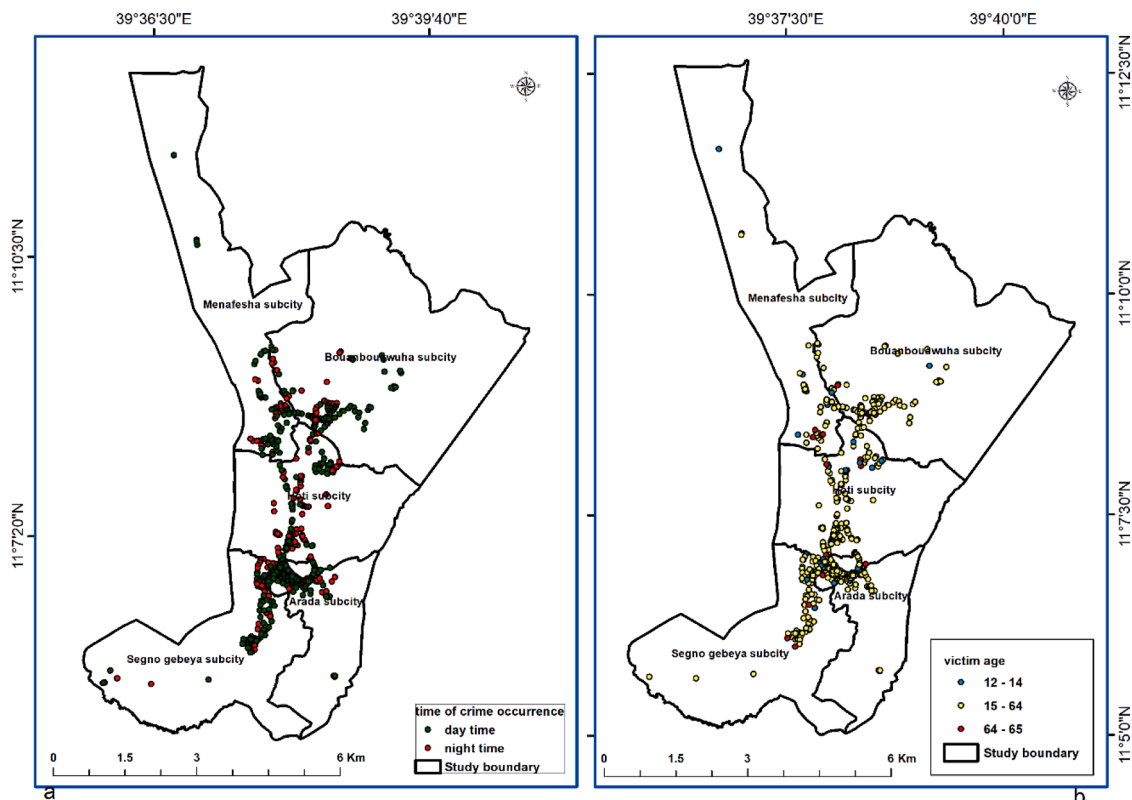


Fig. 5. Temporal distribution of crime incidents(a) and victim age demographics(b) in Dessie city 2018–2021.

the day and at night, as shown in Fig. 5a. Overall, theft represented the majority of crimes (42.14 %), indicating that larceny occurs more frequently during daytime hours. Intimidation incidents occurred at similar rates during the day (12.57 %) and night (12.44 %), suggesting relatively equal temporal distribution. In general, daytime crimes—including theft and burglary—were more prevalent than nighttime offenses, whereas murders were slightly more frequent at night. Across all crime types, approximately 30 % of incidents occurred at night, while over 70 % took place during the day.

In demographic crime hotspots, victimization is often influenced by age, sex, or other demographic characteristics, as highlighted by the Dessie City Police Department [46–48]. In Ethiopia, individuals are categorized into three age groups: young (0–14 years), working age (15–64 years), and elderly (65 years and over) [49]. Using these classifications, the study mapped crime incidents by age and sex (Fig. 5b). Among the young age group, females constituted 2.73 % of victims, while males accounted for 0.14 %. In the working-age group, females represented 16.07 % and males 25.46 % of victims. Among the elderly, females accounted for 1.28 %, and males for 54.32 % of victims. Notably, the elderly experienced the highest victimization rate, comprising 55.60 % of total victims, likely due to age-related vulnerabilities that increase susceptibility to exploitation and fraud. The working-age group accounted for 41.53 % of cases, and together with the elderly, these two groups represented 97.13 % of all victims, whereas the young age group accounted for only 2.87 %.

The application of ArcGIS Pro Hot Spot Analysis enables the identification of spatial clusters with statistically significant high or low attribute values, particularly regarding criminal incidents. By analyzing weighted data points, such as the total number of crimes per census block, analysts can pinpoint areas with elevated crime rates, referred to as hotspots. This approach not only highlights clusters of high criminal activity but also identifies regions with unexpectedly low crime occurrences, providing insights into neighborhood characteristics or policies that may deter crime [50,51]. The Hot Spot Analysis tool calculates the Getis-Ord G_i^* statistic, producing z-scores and p-values to indicate the significance of clusters. Statistically significant positive z-scores reveal intense clustering of high values (hotspots), whereas significant negative z-scores indicate clustering of low values (cold spots) [52].

In this study, Hote, Arada, and Segno were identified as statistically significant hotspots, reflecting concentrated criminal activity. Conversely, Menafesha and Bounbouwha, despite modest negative z-scores, emerged as significant cold spots, indicating areas with lower-than-expected crime rates. Fig. 6 illustrates the spatial distribution of

these hotspots and cold spots within Dessie, emphasizing areas with statistically significant concentrations of criminal activity that are critical for targeted interventions [53]. The identification of Hote, Arada, and Segno as hotspots also reflects their socio-economic context. These areas function as central markets, transport junctions, and public gathering points, attracting high daily movement of people and goods. According to routine activity theory, such convergence points create opportunities for theft and assault in the absence of effective guardianship. In contrast, Menafesha and Bounbouwha, identified as cold spots, are predominantly residential neighborhoods with relatively stable populations and stronger social cohesion, consistent with social disorganization theory [54]. These findings suggest that crime patterns in Dessie are closely linked to underlying social and economic realities, in addition to spatial clustering.

Spatial autocorrelation measures the degree to which nearby objects exhibit similar values across a given spatial region. Negative autocorrelation arises when similar values are widely separated, while positive autocorrelation occurs when nearby locations have similar values. According to [55], spatial autocorrelation is critical for assessing the relevance of geographic variables, particularly in understanding how spatial features influence specific objects when relationships among locations exist. Moran's I is a widely used statistical measure to quantify spatial autocorrelation [56]. In ArcGIS Pro, the "Spatial Autocorrelation (Moran's I)" tool effectively calculates this index, with geographic units showing strong positive values when adjacent to similar units and strong negative values when adjacent to dissimilar units. A Moran's I value of 0 indicates neither autocorrelation nor complete randomness among geographic units.

The Moran's I approach assists in identifying hotspots and cold spots. The tool calculates the Moran's I index, along with a z-score and p-value, to determine significance. Our analysis yielded a z-score of 3.297616 (greater than 2.58) and a p-value of 0.000975 ($p < 0.01$), indicating strong positive spatial autocorrelation with a Moran's I value of 0.027492. This result demonstrates that crime incidents were clustered with 99 % confidence, as shown in Fig. 7a. As noted by The investigation conducted by Caplan et al. underscores the importance of identifying spatial autocorrelation within crime data. The derived model, expressed as $0.76801\text{Nugget} + 0.28899\text{Stable}$ (136.32, 1.216) as illustrated in Fig. 7b, provides valuable insights into the spatial dynamics of criminal incidents. The presence of a non-zero nugget (0.76801) indicates that variability exists at very short distances that is not captured by spatial correlation. This variability may arise from measurement errors, random fluctuations, or unobserved factors influencing crime patterns. The significant nugget value suggests that local variations play a crucial role in shaping the overall crime landscape, supporting recent findings that highlight the impact of localized factors on crime distribution [57].

The stable component (0.28899) quantifies the degree of spatial correlation among crime incidents at greater distances, indicating that the likelihood of encountering similar crime types diminishes as one moves away from a crime location. This observation aligns with the concept of crime clustering in urban environments, where crime often manifests in concentrated areas due to various social, economic, and environmental influences [58,59]. Individual data points, represented by red dots, illustrate the observed semivariance at various distances. The distribution of these points offers insights into the variability of the data. The trend indicated by the blue model line suggests a relationship between semivariance and distance; specifically, a gradual decline in semivariance with increasing distance implies that nearby data points are more similar [60]. This downward trend reinforces the idea that nearby incidents tend to be more alike than those farther apart, further supporting the concept of spatial autocorrelation [61]. The identified range (136.32) specifies the distance over which spatial correlation significantly affects crime data, suggesting that points within this range are likely to exhibit similar characteristics. Existing literature corroborates the notion that the spatial dependence of crime extends over specific distances, influencing resource allocation for policing and

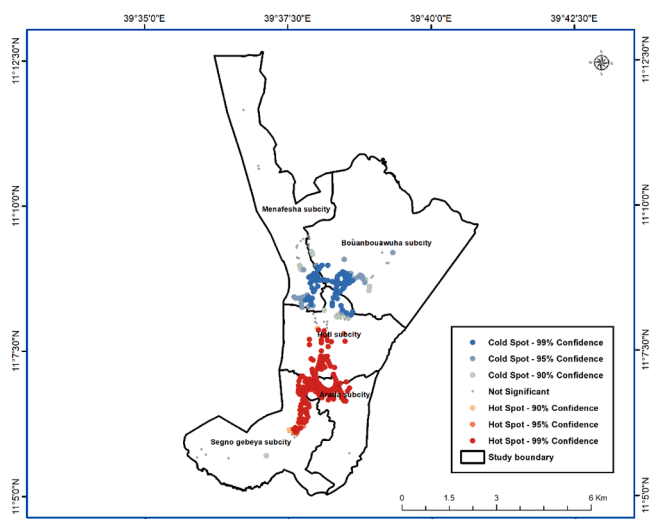


Fig. 6. Map showing statistically significant crime hotspots and cold spots in the study area.

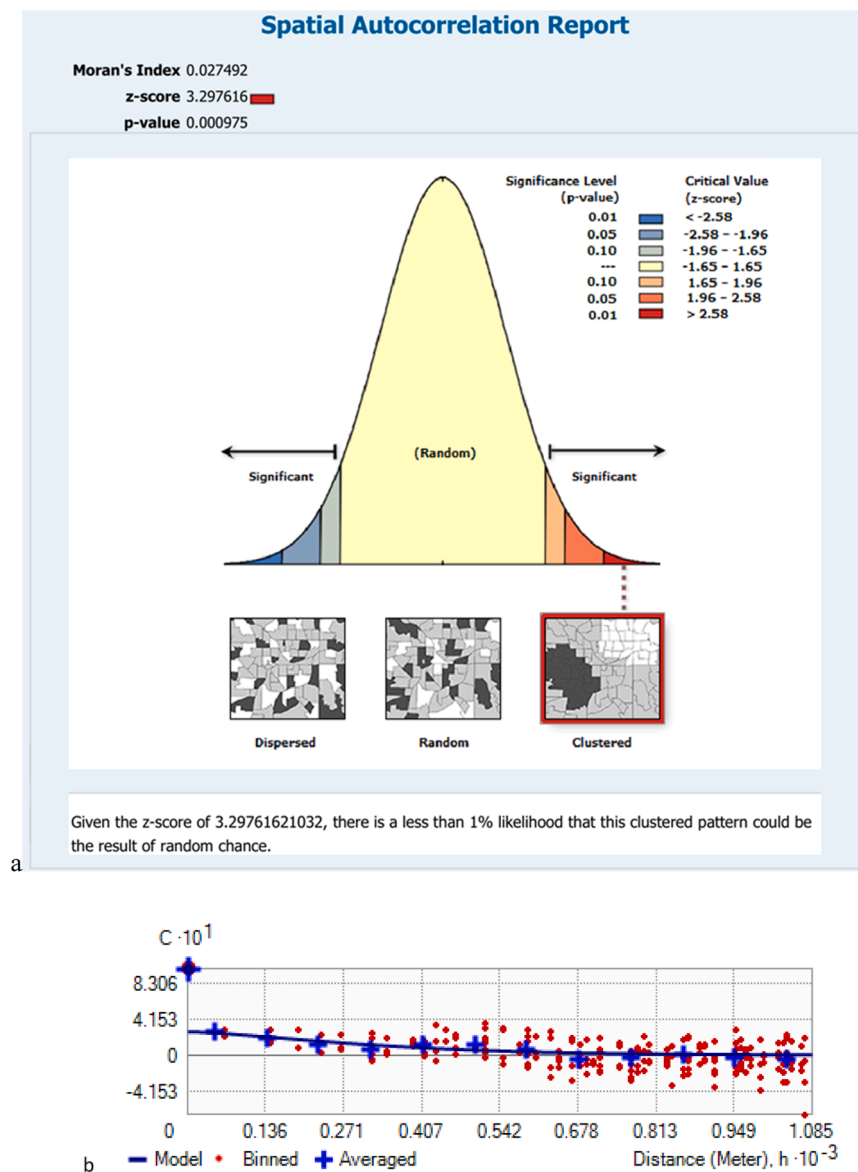


Fig. 7. Spatial autocorrelation analysis(a) and Semi-variogram modeling(b) in the study area.

community safety strategies [62].

The combination of the nugget and stable components contributes to the sill, which represents the overall variance of the variogram. A higher sill indicates pronounced geographic heterogeneity in crime distribution, suggesting that certain areas experience significantly different levels of criminal activity compared to others. Moreover, the high correlation coefficient (0.76801), approaching 1, reinforces the presence of strong spatial associations among crime incidents. This finding aligns with contemporary scholarship advocating for the inclusion of spatial variables in crime analysis to effectively capture underlying patterns. The variogram analysis provides robust evidence of spatial autocorrelation within crime data, highlighting the necessity of incorporating spatial lag or control variables in statistical modeling. Integrating these findings with recent academic discourse underscores the importance of a nuanced understanding of the spatial dynamics of crime for developing effective intervention strategies and enhancing public safety measures, thereby emphasizing the interconnectedness of crime incidents within geographic contexts.

Beyond identifying crime hotspots, crime prediction serves as a valuable tool for both police departments and the broader community [63]. Weighted geographical interpolation was employed to estimate

potential future crime locations, or unidentified hotspots, based on a distance-based algorithm that considers existing hub sites. This technique calculates cell values by linearly weighting nearby sample points. As noted by [64], Inverse Distance Weighting (IDW) assigns higher weights to locations closer to the predicted site, with the influence of each known point decreasing with distance. Fig. 8 illustrates that points nearer to a processing cell exert greater influence than more distant points. Using the currently identified crime hotspots, the IDW analysis predicted potential future crime locations. The interpolated map shows that the northern and southern peripheral areas of the study site, shaded light blue, exhibit lower crime concentrations, while central areas-Hote, Segno Gebeya, and Arada-colored light red and grey, display higher concentrations.

The study also applied the standard deviational ellipse (SDE) to analyze the spatial dispersion and directional trends of crime. The ellipse is defined by the mean center, which is the arithmetic average of all feature locations, and its axes reflect the degree and orientation of dispersion. As indicated by [65], ellipses can effectively represent both the spread and directionality of spatial phenomena. The size and shape of the ellipse capture the extent of dispersion, while its alignment indicates directional trends. Dispersion is measured as the deviation of

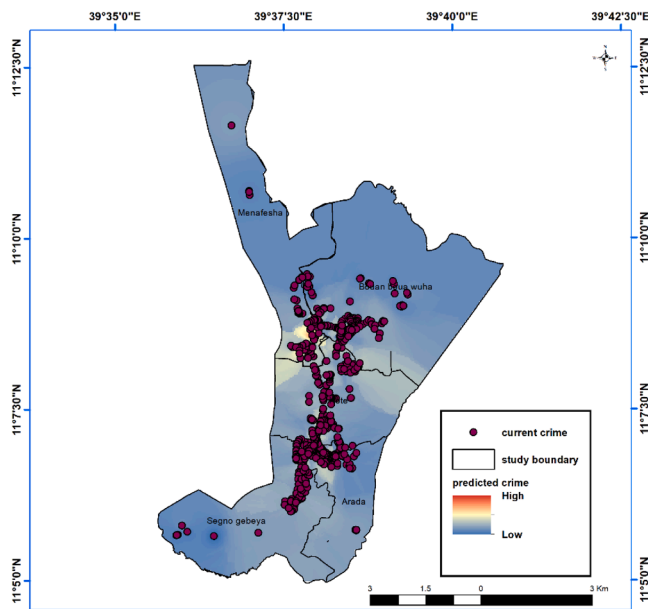


Fig. 8. Future predicted crime based on the current status of the study area.

each feature's location from the mean center, as described by [43]. In this study, the mean center for most crime types was located in Hote (Fig. 9), except for murder, which was concentrated along the southwest boundary of the sub-city. To facilitate rapid offender apprehension, establishing community police stations in these areas is recommended. The SDE analysis revealed a predominantly north-south directional pattern for most crimes, while murder exhibited a north-east-south-west orientation, reflecting a localized variation in spatial concentration.

4. Discussion

This study employed Optimized Hot Spot spatial statistical tools in ArcGIS Pro to generate outputs including z-scores, p-values, and Gi Bin confidence levels, with direct implications for policy and crime prevention. In identified hotspots such as Arada and Hote, police agencies can prioritize community policing strategies to enhance cooperation with residents [47]. The concentration of crime in Hote, Arada, and Segno reflects international evidence that a small proportion of urban

“micro-places” generate disproportionate levels of crime [43,66]. These areas function as transit hubs and commercial centers, where opportunity structures are dense, consistent with situational crime prevention perspectives [67]. Weak informal guardianship, particularly in areas with high daily flows of people, further amplifies risks, echoing social disorganization theory as discussed in the earlier. Studies in Nairobi, Lagos, and Addis Ababa report similar hotspot dynamics, shaped by socio-economic pressures and rapid urbanization [14,68]. The findings suggest that Dessie's crime hotspots cannot be understood purely as statistical clusters but as socially produced spaces where opportunity, motivation, and limited guardianship converge. Thus, hotspot policing strategies in Dessie should be combined with community-oriented interventions, aligning with global evidence on the effectiveness of place-based crime prevention [69]. Situational crime prevention measures such as improved street lighting, formal market regulation, and CCTV surveillance can reduce opportunities for theft and intimidation, aligning with environmental criminology, which emphasizes modifying environments to reduce criminal opportunities. These strategies also support the Amhara Police Commission's focus on community-based policing (2020) while providing precise spatial guidance for resource allocation. Various geographic statistical approaches can identify hot and cold spots [27]. Features within bins indicate confidence levels, with a p-value of 0.01 representing 90 % confidence, while Gi Bin values of zero indicate non-significant clustering. Statistically significant features must be high-value points surrounded by other high values (hotspots) or low-value points surrounded by other low values (cold spots), exceeding patterns expected by chance. ArcGIS Pro's Hot Spot Analysis calculates z-scores for each feature [70], with higher positive z-scores denoting intense clusters of high values and significant negative z-scores indicating clusters of low values.

In this study, high z-scores (0.037 to 4.608) identified statistically significant hotspots in Segno, Arada, and Hote. Factors contributing to these concentrations include Hote's central location, bus station, and open spaces, Arada's Central Business District, and Segno's proximity to major thoroughfares. Conversely, cold spots with low z-scores (−3.231 to −0.116) were observed in Menafesha and Bounbouwha, indicating significantly lower crime activity. Recent geospatial analyses in other Ethiopian cities, such as Mekelle [71], demonstrate the utility of GIS in crime mapping; this study applies similar methods to Dessie, enhancing spatial data integration and crime pattern analysis. Community policing has been emphasized in previous studies [37,72], highlighting the importance of local engagement in crime reduction. This research extends those insights by using geospatial analysis to propose targeted strategies. Environmental criminology, integrating routine activity, crime pattern, and rational choice theories [73], explains how the spatial and temporal distribution of offenders, targets, and opportunities shapes crime patterns. Research suggests that approximately 5 % of street segments, or “micro-places,” account for over 50 % of crimes [74–76], a finding supported by this study, which shows concentrated crime in the south-central areas of Dessie. Previous studies often relied solely on police data [77], limiting spatial context, whereas this study combines comprehensive geospatial analysis with statistical rigor to improve pattern identification.

Identified crime-prone areas include market centers, open spaces, narrow streets, and high-density intersections, particularly affecting working-age and low-income populations. The presence of drugs and alcohol further correlates with higher crime rates. Standard deviation ellipse analysis revealed a predominant north-south crime trajectory, with murder exhibiting a northeast-southwest orientation. Theft, the most frequent crime, predominantly occurred during the daytime, consistent with prior research in Dessie and similar urban settings [9,54, 78]. The hotspot identification of Hote, Arada, and Segno aligns with environmental criminology theories, which attribute crime concentration to opportunity structures and social disorganization [35,79]. These findings underscore the necessity for tailored law enforcement strategies targeting identified hotspots. Geospatial analysis provides actionable

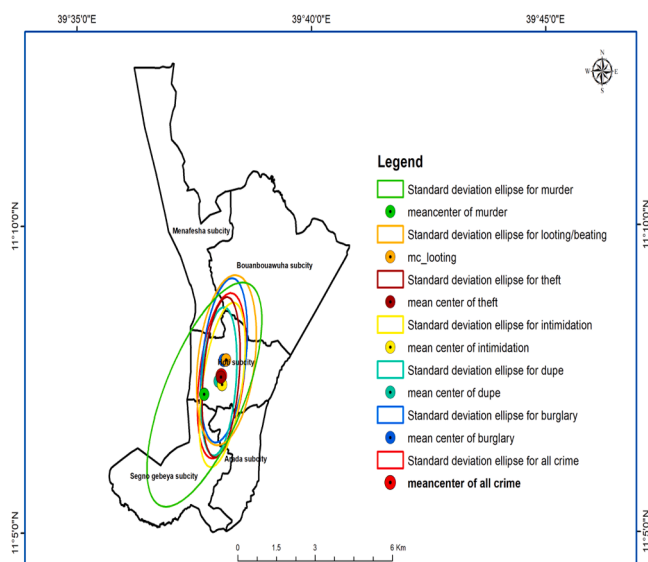


Fig. 9. Standard deviation ellipse with mean center crime in the study area.

insights for policymakers to deploy resources efficiently, reducing crime in high-risk areas [80]. Targeted interventions in crime hotspots have been shown to significantly reduce offenses, particularly in urban environments [81]. Beyond technical contributions, this study links socio-economic vulnerabilities and routine activities to urban geography, advancing both spatial analysis methods and criminological understanding. By integrating geospatial insights with established crime theories, the findings offer practical guidance for crime prevention and enrich the academic discourse on urban crime in Ethiopia, ensuring relevance for rapidly urbanizing cities such as Dessie.

5. Conclusion

This study has demonstrated how geospatial techniques can provide a deeper understanding of crime distribution and dynamics in Dessie City. By applying hotspot analysis, Moran's I, and semivariogram modeling, the research confirmed that crime is not randomly distributed but exhibits strong clustering. The neighborhoods of Hote, Arada, and Segno were identified as persistent hotspots, while Menafesha and Bounbouwha showed relatively low levels of crime. These spatial patterns are closely linked to socio-economic realities, with central markets, transport hubs, and open spaces creating opportunities for theft, intimidation, and related offenses. Temporal analysis indicated that most crimes occurred during the daytime, and demographic assessment revealed that elderly and working-age groups are disproportionately affected. Furthermore, directional analysis highlighted a predominant north-south trajectory of crime, with murder showing a distinct northeast-southwest orientation. Beyond its local policy relevance, this study demonstrates how geospatial crime analysis can bridge technical methods with criminological interpretation. Scholars have recently emphasized that spatial statistics should be complemented by socio-theoretical perspectives to avoid reducing crime mapping to descriptive outputs [82–84]. By combining geospatial evidence with criminological theory, the findings from Dessie provide a more comprehensive explanation of why crime concentrates in particular urban settings. This integration strengthens the contribution of geospatial crime studies from the Global South, where empirical research remains limited, and situates Ethiopian urban crime within global criminological debates.

The findings emphasize the need for targeted interventions tailored to hotspot areas, combining community policing with situational prevention measures such as improved street lighting, CCTV surveillance, and regulation of market activities. Integrating geospatial evidence into local policing strategies can support more efficient deployment of scarce resources, reduce crime, and improve public trust. Ultimately, this study underscores the critical role of geospatial technology in advancing data-driven approaches to crime prevention in rapidly urbanizing Ethiopian cities.

CRedit authorship contribution statement

hailu kebede: Writing – original draft, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Mohammed Motuma Assen:** Writing – review & editing, Supervision, Data curation. **Merid Abadi Sharew:** Resources, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Hailu Ayene reports financial support and administrative support were provided by Wollo University

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